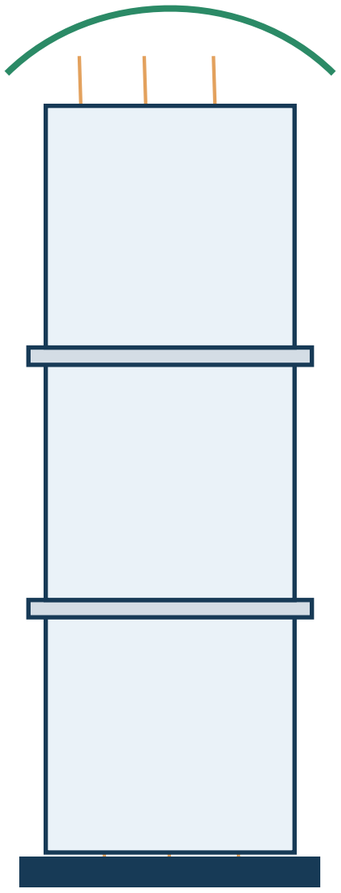
TUBULAR DAYLIGHT TUBE - ENHANCED FABRICATION GUIDE

Readable companion to the 5-sheet prototype fabrication package



DIA559 clear optical bore

Purpose

This edition keeps the original prototype geometry and acceptance limits, but adds plain-language explanations, decision logic, a worked cut-length example, and a shop-ready checklist. It is intended to make the existing drawing set easier to fabricate, inspect, and review.

IMPORTANT - SCOPE HAS NOT CHANGED

The original package is still a prototype tube only. Building supports, fire rating, ceiling conditions, routing, diffuser selection, and project-specific approvals remain open where the source drawings say they are open or on HOLD.

559 mm clear optical bore	**1200 mm** standard module	**>=95%** minimum visible reflectance
1.0 mm max joint gap/step	**1200 mm** max support spacing	**12,000 mm** study max length; photometry pending

How to use this file

- Original sheet pages are preserved. Use them as the dimensional source.
- Explanation pages follow each original sheet. They describe what the dimensions mean and what to check before moving on.
- Red or amber panels are stop/hold conditions. Do not interpret them as fabrication release.
- Worked examples are illustrative only. Replace example dimensions with actual project measurements.

TUBE QUICK REFERENCE

System path, part roles, and non-negotiable prototype constraints

SYSTEM PATH - WHAT EACH PART DOES

B-01	**Existing dome collector base** Transfers daylight into the tube; do not hang tube weight from it.
T-00	**Top adapter, H150** Connects B-01 to the modular tube. Tube enters collar 100 mm.
T-01 / T-01S	**Optical tube modules** Rigid reflective bore; standard 1200 mm, optional short 600 mm.
T-02	**External split sleeve** Keeps butt joints aligned without hardware projecting into the optical bore.
T-03	**Final cut module** Project-specific trim piece, preferred/final range 300-1199 mm.
T-04 / T-05 / D-01..03	**Lower terminal + diffuser hardware**

BEFORE FABRICATION - DATA NEEDED FROM PROJECT

- Measured collector-to-ceiling optical length L.
- Horizontal roof-to-diffuser centerline offset X.
- Ceiling construction, opening, trim, diffuser type and light-distribution requirements.
- Approved support structure and seismic category.
- Insulation/vapor-control requirements and fire/smoke ratings.
- Room illuminance target and glare criterion.

CRITICAL DIMENSIONAL CONTROLS

- Clear bore: DIA559 must stay continuous through modules and joints.
- Module end quality: length 1200 +/- 1; square within 0.5 across diameter; roundness max-min ID <= 2.0.
- Joint quality: axial gap <= 1.0 and radial step <= 1.0. No rivet, screw, or clamp point may enter the bore.
- Optical surface: visible reflectance certificate >=95%; 98% is the design target; protect the reflective face from scratches.
- Support: independent hangers at every joint and no more than 1200 mm centers; final seismic/bracing design remains project-specific.

DEFAULT ROUTING RULE

The package is for a straight vertical path. If centerline offset X is more than 25 mm, stop fabrication and create a project-specific ray trace and elbow development. Do not ovalize or kink a straight reflective module to make an offset.

WHY JOINT QUALITY MATTERS

The source design treats every interior step, gap, seam projection, and added bend as an optical penalty. The design intent is therefore a smooth circular bore, flush butt joints, high reflectance, and the fewest possible direction changes.

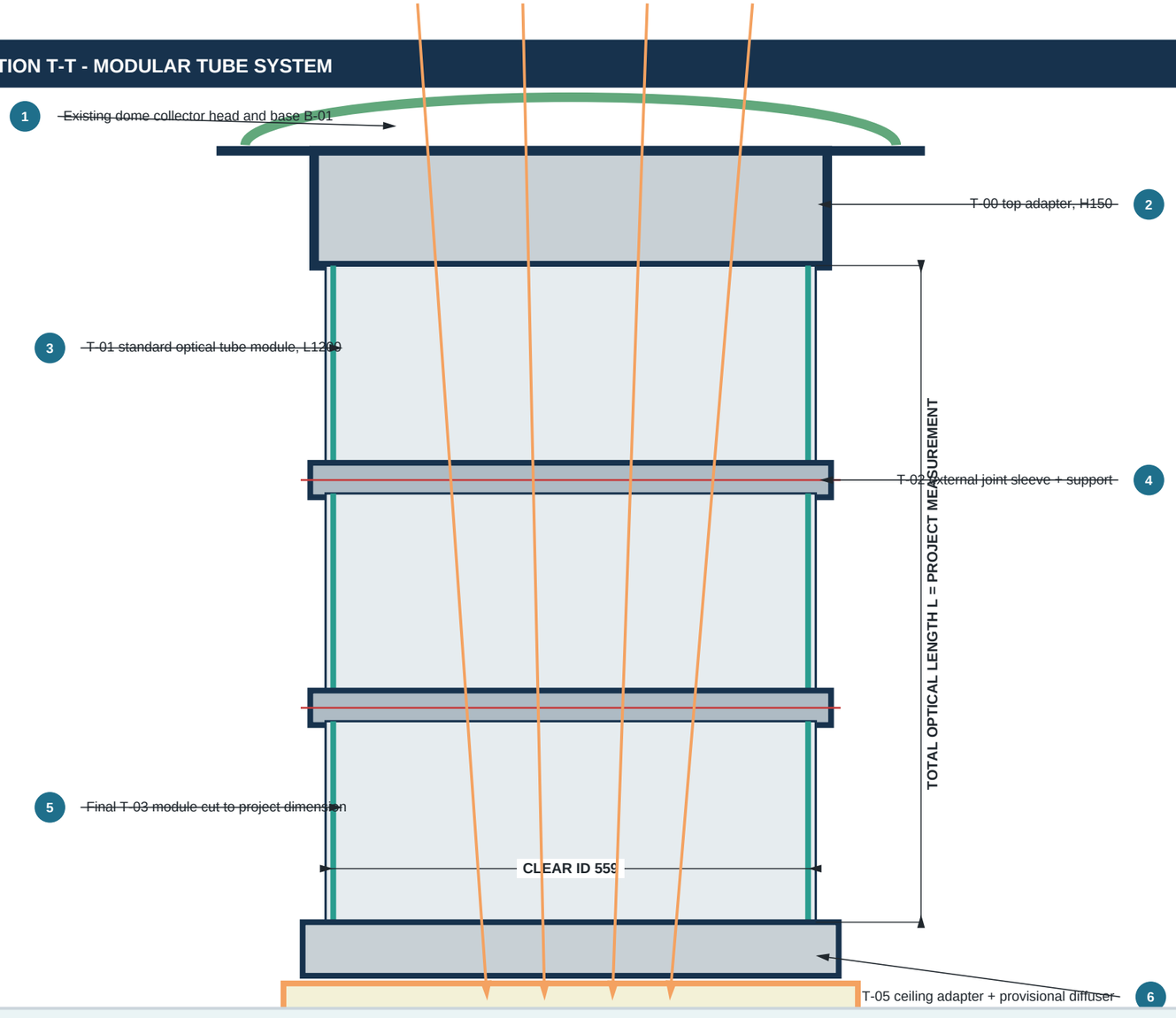
TUBULAR DAYLIGHT TUBE - SYSTEM GENERAL ARRANGEMENT

Modular straight optical tube matched to the dome package: clear bore DIA559, standard module length 1200

DESIGN BASIS: STRAIGHT VERTICAL PATH - TOTAL LENGTH L IS NOT YET KNOWN

Measure roof-collector throat to finished-ceiling interface before cutting the final module. Do not fabricate an offset or elbow from this package.

SECTION T-T - MODULAR TUBE SYSTEM



L = 1200 x N + Ltrim N = number of full modules; use 600 short module if required; preferred Ltrim = 300 to 1199

All module ends butt together: no insertion deduction. Interior step or axial gap at any joint: 1.0 maximum.

FIXED TUBE DESIGN BASIS

PARAMETER	PROTOTYPE VALUE
Clear optical bore	DIA559 continuous
Reflective surface	98% target; 95% minimum visible
Tube material	Enhanced-specular Al, 0.7 nominal
Standard / short length	1200 / 600
Study maximum L	12,000 pending photometry
Preferred routing	Straight and vertical

PROJECT DATA STILL REQUIRED

Measured collector-to-ceiling length L	TBD
Roof and ceiling centerline offset	TBD
Ceiling construction and diffuser type	TBD
Support structure / seismic category	TBD
Required insulation and vapor control	TBD
Fire / smoke rating for tube and insulation	TBD
Room illuminance and glare criteria	TBD

OPTICAL RULES

1. Keep the bore circular and every joint flush; steps scatter light and increase loss.
2. Use the highest practical reflectance. Loss compounds after every wall reflection.
3. Keep longitudinal seams aligned and smooth; place seams away from the dominant low-angle sun side.
4. Support the tube independently. Do not hang its weight from the dome base, roof membrane or ceiling grid.
5. Do not use flexible corrugated duct: its surface and geometry cause severe optical loss.

SHEET 1 EXPLAINED - SYSTEM GENERAL ARRANGEMENT

How to determine tube quantity and what must be known before release

1. LENGTH LOGIC

The general arrangement defines the optical path as a stack of full modules plus one project-cut final module.

$$L = 1200 \times N + Ltrim$$

Where L is the project-measured optical length, N is the number of full 1200 mm modules, and Ltrim is the remaining final cut length. A 600 mm short module can be used when needed to keep the final cut within the preferred range.

JOINT ASSUMPTION

Module ends butt together. The source explicitly states there is no insertion deduction between standard modules.

2. RELEASE GATES ON SHEET 1

- Do not guess total length. Measure from collector throat to finished-ceiling interface before cutting T-03.
- Do not fabricate an elbow from this package. The straight-path assumption is part of the design basis.
- Support and building interfaces are separate approvals. Tube weight must not be carried by the dome base, membrane, or ceiling grid.
- Photometry remains required. The study maximum is 12 m, but final delivered-light performance is not released by length alone.

SHOP / FIELD INPUT FORM

Measured collector-to-ceiling L — TBD / PROJECT INPUT

Roof-to-ceiling centerline offset X — TBD / PROJECT INPUT

Ceiling type / opening / trim — TBD / PROJECT INPUT

Support structure / seismic category — TBD / PROJECT INPUT

Insulation + vapor-control requirement — TBD / PROJECT INPUT

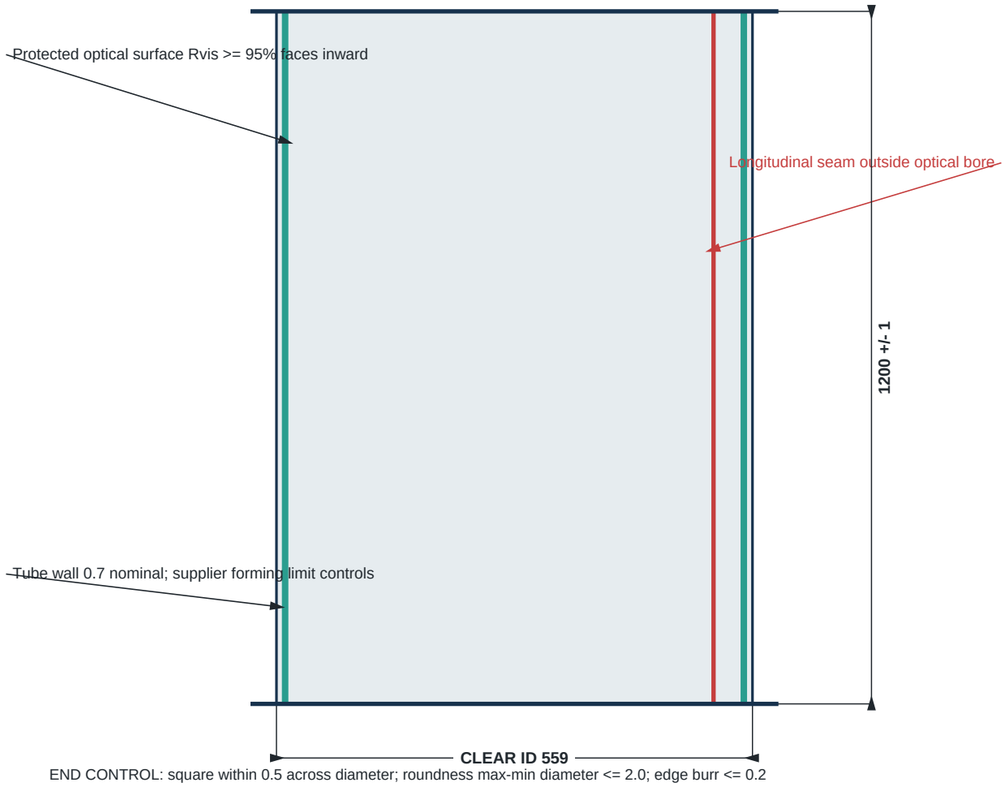
Fire / smoke requirement — TBD / PROJECT INPUT

Target lux + glare criterion — TBD / PROJECT INPUT

STRAIGHT OPTICAL TUBE MODULE - BLANK, ROLLING AND SEAM

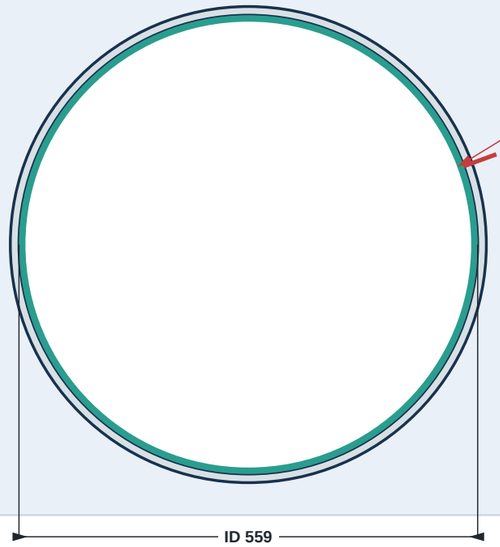
Part T-01: DIA559 clear bore x 1200 long; enhanced-specular aluminum, reflective face inward

PART T-01 - FINISHED MODULE



END CONTROL: square within 0.5 across diameter; roundness max-min diameter <= 2.0; edge burr <= 0.2

SECTION A-A - BORE AND LONGITUDINAL SEAM



ID at seam may reduce by 0.5 max

No rivet or screw may enter the bore

Reflective face scratch limit: 0.25 wide

Finished OD about 560.4 excluding seam

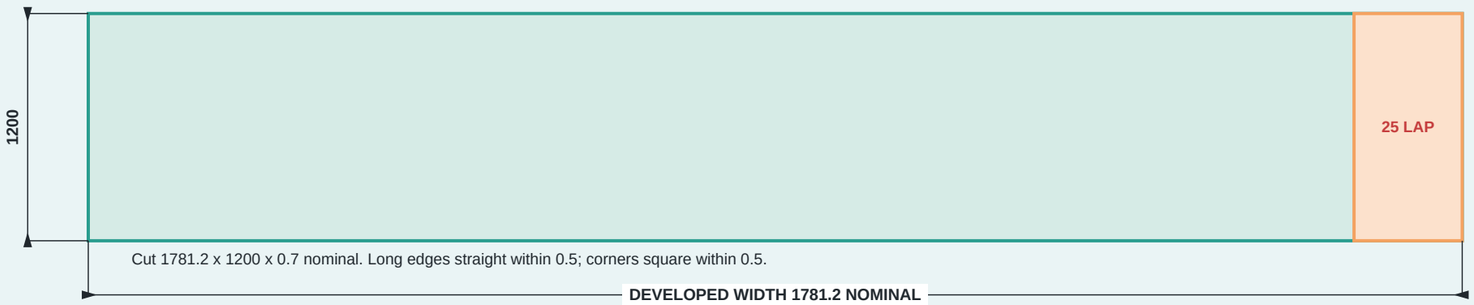
Module mass depends on supplied coil grade

Inspect ID at 0 / 45 / 90 / 135 deg

FLAT BLANK T-01-B - CUT BEFORE ROLLING

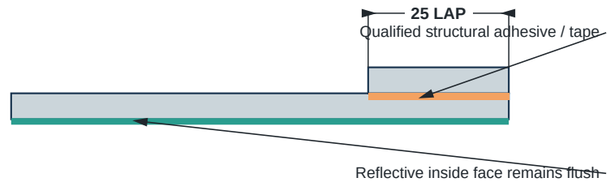
Calculation: pi x 559 clear circumference + 25 lap = 1781.2. Calibrate +/- 2 after the first mandrel trial.

Grain / coil direction runs parallel to tube axis. Keep protective film on optical face until seam closure.



Cut 1781.2 x 1200 x 0.7 nominal. Long edges straight within 0.5; corners square within 0.5.

DETAIL A - EXTERNAL BONDED LAP



1. Clean and prepare only by optical-sheet supplier method.
2. Roll over a DIA559 mandrel with optical face protected.
3. Bond 25 lap on the outside; use a full-length pressure bar.
4. Cure while held round. Do not spot-rivet through the bore.
5. Remove film, clean and inspect after all metalwork ends.
6. Make a seam coupon and peel/shear test before production.

ADHESIVE IS NOT YET SPECIFIED

Select after coil coating compatibility, temperature, humidity and fire review.

SHEET 2 EXPLAINED - T-01 MODULE FABRICATION

Blank size, rolling sequence, seam control, and inspection logic

FABRICATION SEQUENCE - ONE T-01 MODULE

1. **Cut blank**
   1781.2 x 1200 x 0.7 nominal. Keep long edges straight and corners square within 0.5.
2. **Protect optical face**
   Keep protective film on the reflective face until seam closure and metalwork are complete.
3. **Roll**
   Form over a DIA559 mandrel with the reflective face inward.
4. **Make external lap**
   Use a 25 mm outside lap so the optical interior remains flush.
5. **Bond + cure round**
   Use a qualified compatible adhesive/tape and a full-length pressure bar; cure while held round.
6. **Final clean + inspect**
   Remove film only after metalwork ends; verify ID, roundness, seam, scratches, and end squareness.

WHY THE BLANK IS 1781.2 mm WIDE

The source blank development is the clear-bore circumference plus the external lap:

pi x 559 + 25 = 1781.2 mm nominal

The drawing also requires a first-mandrel trial and allows the developed width to be calibrated by +/- 2 mm after that trial. This is important because actual forming behavior depends on the supplied coil and seam process.

ADHESIVE HOLD

No adhesive is released by the source package. Selection must follow coating compatibility, temperature, humidity, and fire review. A seam coupon and peel/shear test is required before production.

ACCEPTANCE CHECKS - T-01

LENGTH 1200 +/- 1 mm	CLEAR ID 559 +/- 1 mm	SEAM ID LOSS <= 0.5 mm	ROUNDNESS max ID - min ID <= 2.0 mm
END SQUARENESS <= 0.5 mm across diameter	BURR <= 0.2 mm	REFLECTANCE certificate >=95% visible	SCRATCH LIMIT 0.25 mm wide noted on reflective face

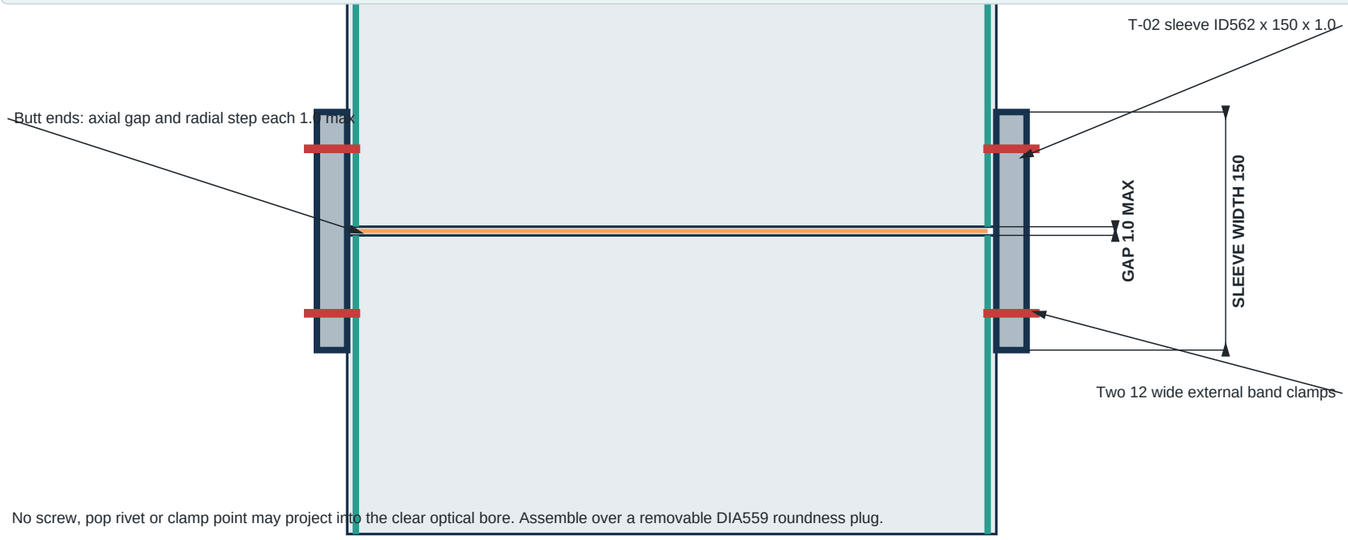
TUBE JOINT, TOP ADAPTER, SUPPORT AND INSULATION

Non-penetrating butt joints preserve the DIA559 optical bore; supports remain independent of roof and ceiling finishes

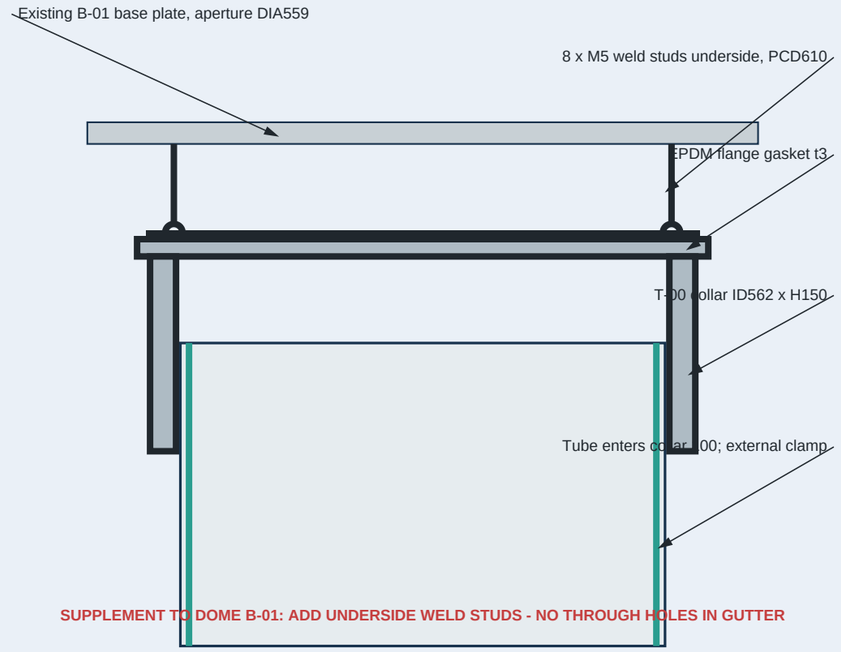
DETAIL J-1 - BUTT JOINT AND EXTERNAL SPLIT SLEEVE

T-02 BLANK: 1795 x 150 x 1.0 aluminum; 30 lap; roll to ID562

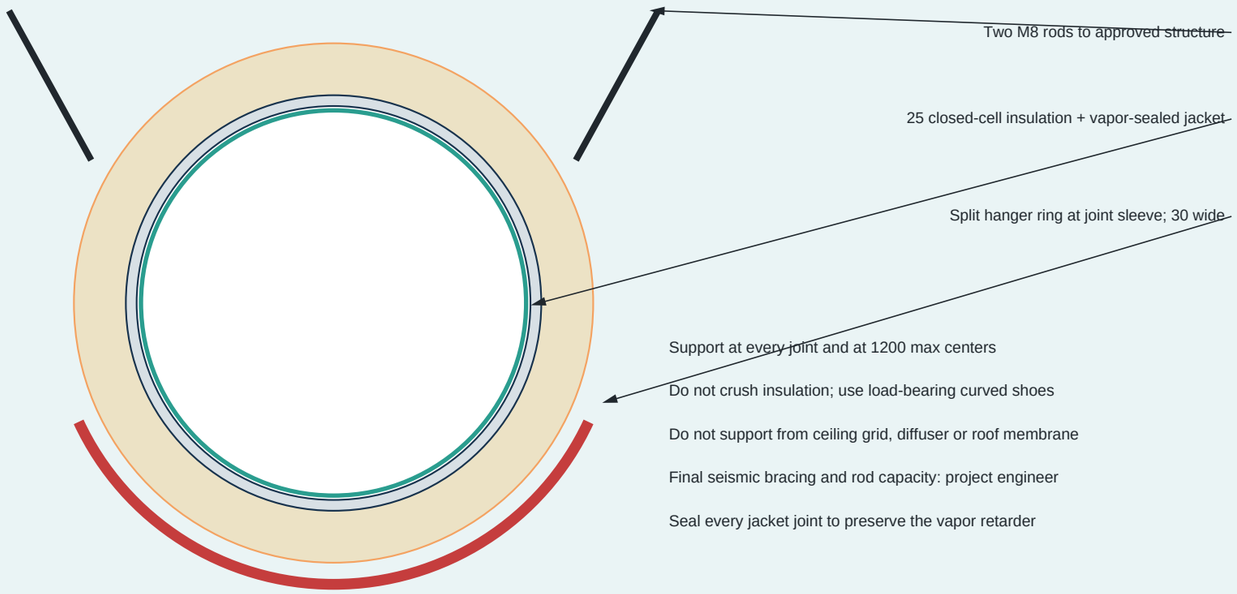
Developed width = pi x 562 + 30 = 1795.4 nominal. Slot the outer lap for clamp access; deburr all edges.



DETAIL J-2 - TOP ADAPTER TO DOME BASE B-01



DETAIL S-1 - INDEPENDENT SUPPORT AT EACH MODULE JOINT



INSULATION / CONDENSATION CONTROL

1. Tube metal is highly conductive; uninsulated attic sections can condense interior moisture.
2. Prototype basis: 25 closed-cell insulation, continuous vapor-sealed jacket, approximate OD612.
3. Insulation type, thickness, flame spread, smoke development and thermal rating remain project-specific.
4. Seal the top adapter and lower terminal against air leakage, but keep the dome gutter drains open.
5. Provide inspection access at the first joint and at the lower terminal.
6. Do not place absorbent insulation inside the optical bore.

SHEET 3 EXPLAINED - JOINTS, SUPPORTS, AND INSULATION

How to keep the bore aligned while carrying structural loads independently

JOINT J-1 - WHAT THE SLEEVE DOES

- T-01 modules meet as a butt joint; the optical bore is not telescoped.
- T-02 is an external split sleeve, ID562 x 150 x 1.0, held by two 12 mm external band clamps.
- Assemble over a removable DIA559 roundness plug so the joint stays circular and aligned.
- Accept only when both axial gap and radial step are <= 1.0 mm.
- No screw, rivet, or clamp point may project into the clear optical bore.

SUPPORT S-1 - LOAD PATH

- Use a split hanger ring at the joint sleeve and two M8 rods to approved structure.
- Provide support at every joint and at no more than 1200 mm centers.
- Use load-bearing curved shoes so support pressure does not crush the insulation.
- Do not support from the ceiling grid, diffuser, roof membrane, or dome base.
- Final rod capacity and seismic bracing are assigned to the project engineer.

TOP ADAPTER T-00

The top adapter connects the modular tube to B-01 while avoiding through-holes in the dome gutter.

- Existing B-01 base aperture: DIA559.
- T-00 collar: ID562 x H150.
- Tube enters collar 100 mm and is externally clamped.
- Supplement to B-01: eight M5 underside weld studs on PCD610.
- Use EPDM flange gasket t3; keep dome gutter drains open.

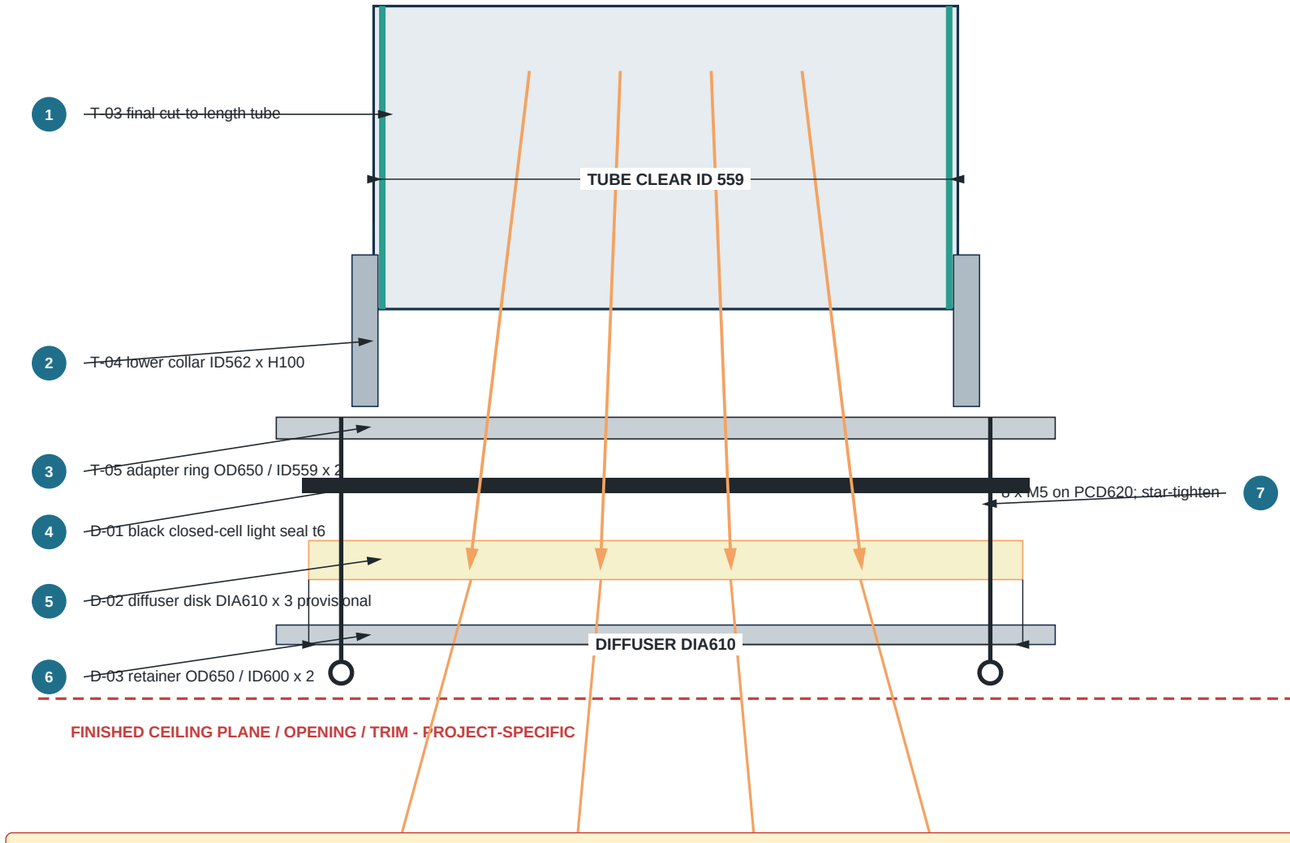
CONDENSATION / VAPOR CONTROL

- Prototype basis is 25 mm closed-cell insulation with a continuous vapor-sealed jacket; approximate insulated OD is 612 mm.
- Seal every jacket joint so the vapor retarder remains continuous.
- Seal the top adapter and lower terminal against air leakage, without blocking gutter drains.
- Provide inspection access at the first joint and lower terminal.
- Insulation type, thickness, flame/smoke and thermal rating remain project-specific.

LOWER TERMINAL, DIFFUSER INTERFACE AND ROUTING CONTROL

Ceiling hardware is provisional until ceiling type, fire criteria and required light distribution are known

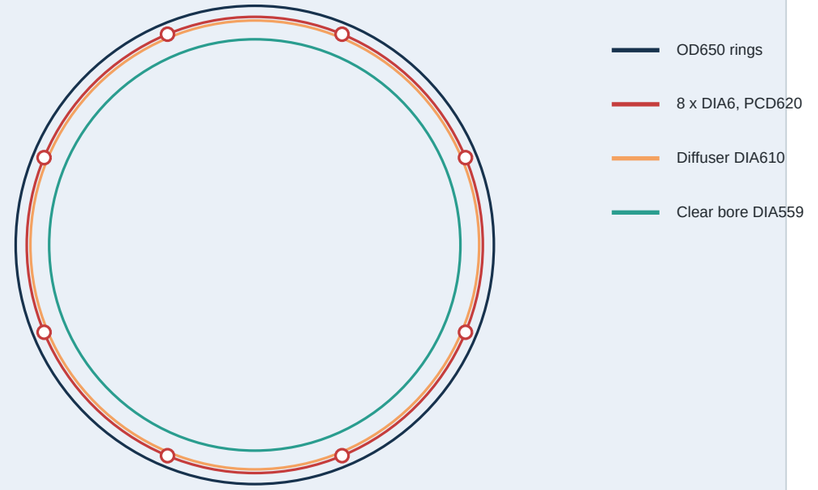
DETAIL D-1 - LOWER TERMINAL EXPLODED SECTION



DIFFUSER PERFORMANCE IS NOT RELEASED

Select optical pattern and transmission after room photometry and glare review. Confirm fire, impact, cleaning and ceiling-plenum requirements before manufacture.

TERMINAL RING - PLAN



OFFSET / ELBOW DECISION - HOLD

DEFAULT: NO ELBOW

1. Measure horizontal offset X between roof and diffuser centerlines.
2. If X <= 25, align straight modules and use terminal trim only.
3. If X > 25, stop fabrication and produce a project ray trace plus elbow development.
4. Do not force, ovalize or kink a straight reflective module to create an offset.
5. Every bend increases reflection count and reduces transmitted light.

CUT-LENGTH WORKSHEET

Measured optical length L = _____ Top adapter occupied length = 100 Lower collar occupied length = 75

Net straight tube length Ltube = L - 175 = _____

Number full 1200 modules N = floor(Ltube / 1200) = _____ Remaining cut length Ltrim = _____

If Ltrim < 300: replace one 1200 module with one 600 module, then recalculate. Minimum final cut module = 300.

Shop release: N x T-01 1200 modules + ____ x T-01S 600 module + one T-03 cut module _____ long.

DIFFUSER RELEASE INPUTS

Room use / target lux	TBD
Ceiling type and opening	TBD
Required transmission / haze	TBD
Glare criterion	TBD
Fire / smoke / impact rating	TBD

SHEET 4 EXPLAINED - LOWER TERMINAL AND CUT LENGTH

Terminal stack, offset decision, and an illustrative length calculation

LOWER TERMINAL STACK - TOP TO BOTTOM

1	**T-03 final cut tube**	Project-sized optical section
2	**T-04 lower collar**	ID562 x H100
3	**T-05 adapter ring**	OD650 / ID559 x 2
4	**D-01 light seal**	Black closed-cell, t6
5	**D-02 diffuser**	DIA610 x 3, provisional
6	**D-03 retainer**	OD650 / ID600 x 2
7	**Fasteners**	8 x M5 on PCD620; star-tighten

OFFSET DECISION

1. **Measure X**
   Horizontal offset between roof and diffuser centerlines.
2. **X <= 25 mm**
   Keep straight modules; handle minor alignment at terminal trim only.
3. **X > 25 mm**
   STOP. Produce project ray trace and elbow development before fabrication.
4. **Never force an offset**
   Do not bend, ovalize, or kink a straight reflective module.

DIFFUSER HOLD

Optical pattern and transmission are not released until room photometry, glare, fire,

OPTICAL CONSEQUENCE

Every bend increases reflection count and therefore reduces transmitted light.

WORKED CUT-LENGTH EXAMPLE - ILLUSTRATIVE ONLY

Assume a measured optical length L = 6000 mm. The source worksheet subtracts 100 mm for the top adapter occupied length and 75 mm for the lower collar.

6000 mm	175 mm	5825 mm	4	1025 mm
Measured L	Less occupied	Net Ltube	Full 1200 modules N	Remaining Ltrim

Result: release 4 x T-01 (1200) plus 1 x T-03 cut to 1025 mm. Because 1025 is within the 300-1199 final-cut range, the 600 mm short module is not needed in this example.

TUBE FABRICATION, OPTICAL CHECK AND RELEASE MATRIX

Manufacturing sequence, bill of materials and tests for the DIA559 modular daylight tube

UNITS: mm unless noted | SCALE: diagrammatic

A. MANUFACTURING AND ASSEMBLY SEQUENCE

01	**CONFIRM ROUTE**	Measure L and centerline offset; release straight layout only when X <= 25.
02	**RECEIVE COIL**	Verify supplier grade, visible reflectance certificate, coating direction and forming limits.
03	**CUT BLANKS**	T-01: 1781.2 x module length; T-02 sleeve: 1795 x 150; protect optical face.
04	**ROLL MODULES**	Use DIA559 mandrel; make external 25 lap; cure round; trim square.
05	**INSPECT**	ID, roundness, length, seam, scratches and reflectance evidence; reject defects.
06	**ADD ADAPTERS**	Install T-00 under dome base and lower T-04/T-05 terminal without bore projections.
07	**JOIN**	Butt modules over alignment plug; fit T-02 split sleeve and two external clamps.
08	**SUPPORT**	Hang at every joint / 1200 max centers from approved structure; add bracing as designed.
09	**INSULATE**	Apply continuous approved insulation and vapor jacket; seal every joint and penetration.
10	**TEST/RELEASE**	Check air, condensation, supports and photometry before enclosing the ceiling.

B. INSPECTION AND TEST MATRIX

ITEM	ACCEPTANCE	STATUS
Module length	1200 +/- 1; cut module per worksheet	OPEN
Clear bore	559 +/- 1; seam reduction <= 0.5	OPEN
Roundness	Max ID - min ID <= 2.0	OPEN
End squareness	0.5 across diameter	OPEN
Joint alignment	Gap and radial step each <= 1.0	OPEN
Optical surface	Rvis cert >=95%; target 98%; no major scratch	OPEN
Seam coupon	Pass qualified peel / shear criterion	HOLD
Air leakage	Project criterion; smoke pencil / pressure test	HOLD
Condensation	No interior wetting in design condition	HOLD
Support / seismic	Approved loads, spacing and bracing	HOLD
Fire / smoke	Tube, adhesive and insulation approved	HOLD
Photometric	Room lux and glare criteria achieved	HOLD
Access / cleaning	Top, first joint and terminal accessible	OPEN

C. APPROXIMATE REFLECTION LOSS - STRAIGHT TUBE

For a simplified central ray: number of wall hits N approximately equals L x tan(theta) / D; surface throughput approximately rho^N.

Example below: L = 6000, D = 559. This excludes dome, joint, dirt and diffuser losses.

RAY ANGLE TO AXIS	N APPROX	AT rho=0.95	AT rho=0.98
15 deg	3	86%	94%
30 deg	6	74%	89%
45 deg	11	57%	80%

DESIGN CONSEQUENCE: 98% reflector material and a straight, clean bore materially improve delivered light.

Use measured photometry for release. This simplified relation is for design comparison, not a product rating.

D. BILL OF MATERIALS AND TECHNICAL BASIS

PART	QTY	DESCRIPTION
T-00	1	Top collar ID562 x H150 + OD640 flange
T-01	N	Optical tube ID559 x 1200 x 0.7 nominal
T-01S	opt	Optical tube ID559 x 600 x 0.7 nominal
T-02	N	External split sleeve ID562 x 150 x 1.0
T-03	1	Final optical tube, cut length 300-1199
T-04	1	Lower collar ID562 x H100
T-05	1	Terminal adapter OD650 / ID559 x 2
D-01	1	Closed-cell light seal t6
D-02	1	Diffuser DIA610 x 3 - PROVISIONAL
D-03	1	Retainer OD650 / ID600 x 2
S-01	N+1	Split support ring + two M8 hanger rods
I-01	L	Approved insulation / vapor jacket - TBD

Basis: Alanod daylight reflector surfaces - visible reflection up to 98%; coil and cut-sheet formats.

Basis: VELUX commercial 22 in tubular daylight devices - high-reflectance rigid tube and modular connections.

Basis: FGIA NAFS - TDDCC and TDDOC classifications; adopted edition and performance criteria are project-specific.

This is a generic prototype design and does not reproduce a proprietary connector, elbow or diffuser.

SHOP TRAVELER + RELEASE CHECKLIST

A consolidated sequence based on Sheets 1-5

A. SHOP TRAVELER

1. **Confirm route**
   Record L and X. Straight layout only when X <= 25.
2. **Receive coil**
   Verify grade, reflectance certificate, coating direction, forming limits.
3. **Cut blanks**
   T-01 width 1781.2; T-02 sleeve 1795 x 150; protect optical face.
4. **Roll modules**
   Use DIA559 mandrel; external 25 lap; cure round; trim square.
5. **Inspect**
   ID, roundness, length, seam, scratches, reflectance evidence.
6. **Add adapters**
   Fit T-00 and lower T-04/T-05 without bore projections.
7. **Join**
   Butt over alignment plug; T-02 sleeve + two external clamps.
8. **Support**
   Every joint / 1200 max centers from approved structure.
9. **Insulate**
   Continuous approved insulation and vapor jacket; seal joints.
10. **Test / release**
    Check air, condensation, supports, access and photometry before ceiling close-in.

B. OPEN VS HOLD

OPEN - verify and document

- Module dimensions / clear bore
- Roundness / end squareness
- Joint alignment
- Reflectance evidence
- Access / cleaning

HOLD - project criterion needed

- Seam coupon criterion
- Air leakage criterion
- Condensation design condition
- Support / seismic approval
- Fire / smoke approval
- Room photometry / glare

C. WHY REFLECTANCE MATTERS

The source provides a simplified straight-tube comparison for L = 6000 mm and D = 559 mm. More wall hits amplify the benefit of a higher-reflectance surface.

Ray angle	Hits N	rho=0.95	rho=0.98
15 deg	3	86%	94%
30 deg	6	74%	89%
45 deg	11	57%	80%

RELEASE RULE

Use measured photometry for release. The simplified reflection relation is for design comparison, not a product rating.

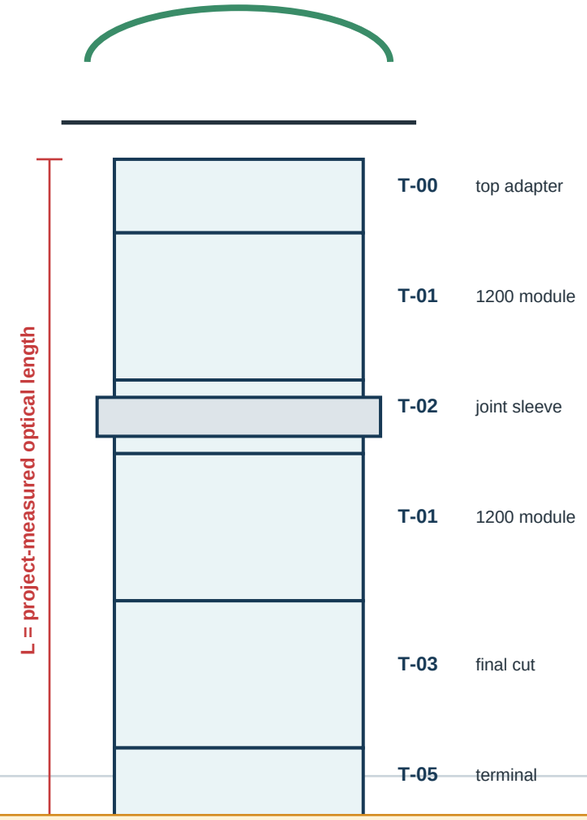
TUBE TERMINOLOGY, DIMENSIONS & INTERFACES

A plain-language key to the dimensions and part codes already used in the source package

A. SYSTEM TERMS - WHAT EACH SYMBOL MEANS

TERM	MEANING	SOURCE RULE / LIMIT
L	Measured optical length from collector throat to finished-ceiling interface	Must be measured before the final module is cut.
Ltube	Net straight tube length used by the cut worksheet	Ltube = L - 175 mm.
N	Number of full 1200 mm modules	N = floor(Ltube / 1200).
Ltrim	Remaining length assigned to final T-03 module	Preferred/final range 300-1199 mm.
X	Horizontal offset between roof and diffuser centerlines	X <= 25 mm: straight layout. X > 25 mm: stop and create project-specific ray trace/elbow development.
ID / OD	Inside / outside diameter	Clear optical ID is 559 mm; sleeve and collars are outside the optical bore.
Rvis	Visible reflectance of the optical surface	Certificate >=95%; design target 98%.

B. PART RELATIONSHIPS - FROM ROOF TO ROOM



- B-01 is the existing collector base; it transfers light but must not carry the tube weight.
- T-00 connects B-01 to the tube; the tube enters the collar 100 mm.
- T-01 / T-01S form the rigid reflective path; standard 1200 mm and optional 600 mm modules.
- T-02 aligns each butt joint externally; nothing may protrude into the DIA559 optical bore.
- T-03 is the only project-cut optical module in the release worksheet.
- T-04 / T-05 / D-01..03 form the lower terminal; diffuser selection remains provisional.

READ THIS BEFORE USING THE DIMENSIONS

The source uses diagrammatic drawings, so written dimensions, formulas, acceptance limits, OPEN/HOLD status and project-input fields control over apparent graphic scale.

C. FIVE NON-NEGOTIABLE OPTICAL / ASSEMBLY IDEAS

1	Continuous circular bore	Keep DIA559 clear and circular through modules and joints.
2	Flush joints	Axial gap and radial step are each 1.0 mm maximum.
3	No bore intrusions	No screw, rivet or clamp point may enter the clear optical bore.
4	Independent support	Do not hang tube weight from dome base, roof membrane, ceiling grid or diffuser.
5	Straight path by default	Do not kink, ovalize or fabricate an elbow from the straight-module package.

DETAILED TUBE FABRICATION FLOW

The same source sequence expanded into inputs, actions, checks and stop conditions

01 CONFIRM ROUTE

ACTION Measure L and X.
CHECK L known; straight path released only when X <= 25.
HOLD STOP if X > 25.

02 CONFIRM PROJECT INPUTS

ACTION Capture ceiling, support, insulation, fire/smoke and lighting requirements.
CHECK Open items are visible before shop release.
HOLD Do not treat TBD fields as approved.

03 RECEIVE OPTICAL COIL

ACTION Verify grade, reflectance certificate, coating direction and forming limits.
CHECK Rvis evidence >=95%.
HOLD Reject missing/insufficient evidence.

04 CUT BLANKS

ACTION T-01 width 1781.2 nominal; protect reflective face.
CHECK Correct blank size and clean edges.
HOLD Calibrate width after first mandrel trial.

05 ROLL + BOND MODULES

ACTION Roll on DIA559 mandrel; external 25 mm lap; cure round.
CHECK Clear bore remains smooth and round.
HOLD Adhesive remains a HOLD until compatibility/fire review.

06 INSPECT MODULES

ACTION Check ID, length, roundness, end squareness, seam and scratches.
CHECK All OPEN dimensional items documented.
HOLD Reject outside acceptance limits.

07 JOIN MODULES

ACTION Butt ends; alignment plug; T-02 sleeve + two external clamps.
CHECK Gap <=1.0; radial step <=1.0; no bore projection.
HOLD Correct before insulation.

08 INSTALL ADAPTERS

ACTION Install T-00, lower T-04/T-05 without bore projections.
CHECK Interfaces align with DIA559 path.
HOLD Keep dome gutter drains open.

09 SUPPORT

ACTION Hang at every joint / 1200 mm max centers from approved structure.
CHECK Tube load independent of roof/ceiling finishes.
HOLD Final seismic/bracing is project engineer scope.

10 INSULATE + VAPOR SEAL

ACTION Apply approved continuous insulation and seal jacket joints.
CHECK No crushed insulation; vapor retarder continuous.
HOLD Material/fire properties remain project-specific.

11 TERMINAL + DIFFUSER

ACTION Assemble provisional diffuser interface.
CHECK Terminal concentric; access maintained.
HOLD Diffuser performance is not released until photometry/glare review.

12 TEST / RELEASE

ACTION Check air, condensation, supports and photometry before closing ceiling.
CHECK OPEN items complete; HOLD items closed by project approval.
HOLD No final release with unresolved HOLD items.

TUBE TOLERANCE & INSPECTION CONTROL MAP

One place to see each quantitative acceptance limit and its fabrication meaning

CONTROL ITEM	SOURCE ACCEPTANCE	WHEN TO VERIFY	WHY IT MATTERS / RESPONSE IF OUTSIDE LIMIT
Module length	1200 +/- 1 mm; cut module per worksheet	After trimming, before joining	Stack length and terminal position depend on module length. Correct or reject before assembly.
Clear bore	559 +/- 1 mm; seam reduction <=0.5 mm	After rolling and seam cure	Maintains optical area and fit with alignment/joint system.
Roundness	Max ID - min ID <= 2.0 mm	After cure; also before final joining	Excess ovality can create steps and poor sleeve fit.
End squareness	0.5 mm across diameter	After end trim	Unsquare ends can create joint gaps/steps even when the sleeve is fitted.
Joint alignment	Axial gap <=1.0 mm; radial step <=1.0 mm	At every butt joint before insulation	Interior discontinuities add optical loss; re-align before locking the joint.
Optical surface	Rvis certificate >=95%; target 98%; no major scratch	Incoming material and final visual inspection	Reflectance loss compounds with repeated wall reflections. Protect/clean only as allowed by material supplier.
Seam scratch limit	0.25 mm wide	After film removal and final cleaning	Source sets a specific scratch limit at the reflective face.
Support spacing	At every joint and 1200 mm max centers	Before insulation and ceiling closure	Tube must be independently supported; final loads/bracing remain project-specific.
Final cut module	300-1199 mm preferred/final range	Cut-length release	If remainder is under 300 mm, use one 600 mm short module and recalculate.

INSPECTION SEQUENCE THAT AVOIDS REWORK

Verify geometry before the optical surface becomes difficult to access: inspect each module first, then verify every joint while the alignment plug and sleeve are accessible, then complete supports, insulation, lower terminal and final system tests. This expands the ordering already shown in the source manufacturing sequence.

CUT-LENGTH LOGIC - WORKED DECISION EXAMPLES

Illustrative calculations using the exact worksheet rules; replace examples with project measurements

EXAMPLE A - REMAINDER IS ALREADY ACCEPTABLE

Measured optical length L	6000 mm
Less occupied length	100 + 75 = 175 mm
Net straight tube length Ltube	5825 mm
Full 1200 modules	N = floor(5825/1200) = 4
Remaining Ltrim	5825 - 4x1200 = 1025 mm
Release stack	4 x T-01 + 1 x T-03 at 1025 mm

RESULT

Ltrim = 1025 mm is within the 300-1199 mm range, so the 600 mm short module is not needed.

EXAMPLE B - REMAINDER IS TOO SHORT, SO RECALCULATE

Measured optical length L	5000 mm
Net straight tube length Ltube	5000 - 175 = 4825 mm
First calculation	N = 4; remainder = 25 mm
Rule triggered	25 mm < 300 mm
Replace one full module	3 x 1200 + 1 x 600 = 4200 mm
Recalculated final cut	4825 - 4200 = 625 mm
Release stack	3 x T-01 + 1 x T-01S + 1 x T-03 at 625 mm

RESULT

The short-module rule converts an unusably short 25 mm trim into a 625 mm final cut module.

FIELD-TO-SHOP DECISION TREE

1. Measure L

Do not estimate total optical length.

→

2. Compute Ltube

Ltube = L - 175 mm.

→

3. Compute N + Ltrim

N = floor(Ltube/1200).

→

4. Is Ltrim >= 300?

YES: release full modules + T-03. NO: replace one 1200 with one 600 and recalculate.

→

5. Check X

X <=25: straight path. X >25: STOP fabrication for project-specific offset design.

RISK / CONDITION	SOURCE-COMPLIANT CONDITION	WHAT TO CHECK BEFORE MOVING ON
Fastener, rivet or clamp enters bore	Nothing projects into the clear DIA559 optical path.	Inspect the full circumference of each joint and seam before insulation.
Module ends do not meet cleanly	Axial gap <=1.0 mm and radial step <=1.0 mm.	Use the removable DIA559 roundness/alignment plug during joint assembly as specified.
Tube load carried by roof or ceiling finish	Independent hanger ring at joint sleeve with two M8 rods to approved structure.	Confirm support path is structural and not the dome base, roof membrane, ceiling grid or diffuser.
Support deforms insulation	Do not crush insulation; use load-bearing curved shoes.	Check the support interface remains load-bearing without collapsing the insulation layer.
Vapor jacket interrupted	Every jacket joint is sealed to preserve the vapor retarder.	Inspect every joint and penetration before concealment.
Uninsulated conductive metal in attic condition	Prototype basis is 25 mm closed-cell insulation with continuous vapor-sealed jacket.	Final type/thickness/fire ratings remain project-specific, so do not substitute an unapproved system.
Air leakage at top or lower terminal	Seal both interfaces against air leakage while keeping dome gutter drains open.	Check seals and drainage together; do not solve air leakage by blocking drains.
No inspection access	Access is required at first joint and lower terminal.	Confirm access remains after ceiling/building work is closed.

OPTICAL REASON

The source repeatedly treats steps, gaps, seam projections and extra bends as light-loss mechanisms. Mechanical neatness is therefore part of optical performance, not just fit and finish.

CONDENSATION REASON

The source identifies the tube metal as highly conductive and therefore susceptible to interior moisture when attic sections are uninsulated. Final insulation properties remain a project decision.

TUBE RELEASE READINESS & SIGN-OFF WORKSHEET

A consolidated handoff page using only the source OPEN/HOLD items and project inputs

A. PROJECT INPUTS - COMPLETE BEFORE FINAL RELEASE

Measured collector-to-ceiling length L	PROJECT INPUT
Roof / ceiling centerline offset X	PROJECT INPUT
Ceiling construction + diffuser type	PROJECT INPUT
Support structure + seismic category	PROJECT INPUT
Insulation + vapor-control requirement	PROJECT INPUT
Fire / smoke rating for tube and insulation	PROJECT INPUT
Room illuminance + glare criteria	PROJECT INPUT

B. VERIFICATION MATRIX - STATUS TO CLOSE

OPEN	Module length / final cut	HOLD	Seam coupon criterion
OPEN	Clear bore + seam reduction	HOLD	Air leakage project criterion
OPEN	Roundness	HOLD	Condensation design condition
OPEN	End squareness	HOLD	Support / seismic approval
OPEN	Joint alignment	HOLD	Fire / smoke approval
OPEN	Optical surface evidence	HOLD	Photometric room lux + glare
OPEN	Access / cleaning		

C. SHOP RELEASE RECORD

Project / location	**Measured L** mm	**Measured X** mm
Module stack released	**T-03 final cut length** mm	**Reflectance certificate reference**
Seam coupon record	**Support / seismic approval ref.**	**Fire / smoke approval ref.**
Photometric approval ref.	**Inspector / date**	**Final project approval / date**

FINAL RELEASE RULE

Do not interpret completed shop dimensions as building approval. The source keeps support/seismic, fire/smoke, air/condensation criteria and photometric performance on HOLD until project-specific requirements are closed.